Mitchell Gray

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Education

Georgia Tech, Atlanta, GA — *GPA: 4.0*
Master of Science, Computer Science — *Artificial Intelligence*

Cornell University, Ithaca, NY
Bachelor of Science, Computer Science — *Distributed Systems*
Minor, Operations Research Information Engineering — *M.Eng Equivalence*

Focus Areas: Distributed Systems · Systems Programming · AI · Machine Learning · Databases · Cloud

Technical Skills

Languages: C · C++ · Python · Java · Rust · PL/SQL
Software & Tools: Git · Pytorch · Docker · Kubernetes · Scikit-learn · NumPy · Pandas · Poetry · CMake · CI/CD

Relevant Work Experience

Oracle — **Redwood City, CA**
Software Engineer II | **C, PL/SQL, Raft, Java, OracleDB, OCI** — **Sept 2025 – Present**

- Developing Raft Replication infrastructure in the Oracle Globally Distributed Database used for Shard Replication · Writing fault-tolerant and scalable Distributed Systems code in **C**, **Java**, and **PL/SQL**.
- Received **Training & Mentorship Excellence Award** · Provided technical mentorship to our newest engineer.
- Designed and implemented a generic shard-operations marker framework that consolidated multiple ad-hoc code paths into a single framework · Improved maintainability, scalability, and ease of creating new markers with fewer defects.
- Architected and implemented advanced deployment algorithms when specifying server rack requirements to more evenly balance replicas across shards automatically · Improved CLI user-interface to view user-specified rack information.
- Implemented comprehensive Pluggable DBs lifecycle management · Enabled seamless Drop, Unplug, Plug, Clone, and Relocate operations to streamline maintenance for Raft replicas using Oracle's multi-tenant container architecture.
- Led remediation of shard-to-shard TDE encryption defects by unifying divergent serialization paths · Preserved encryption metadata during replica transfer, enabling reliable decryption on receiving shards and restoring correctness for TDE-enabled shard replication workflows.
- Improved slow-replica flow control · Implemented new throttling logic to stabilize replication with degraded replicas.

Software Engineer I — **June 2024 – Sept 2025**

- Engineered a fault-tolerant replica replication feature on Oracle Cloud Infrastructure · Enforced placement of Raft replicas on separate server racks to eliminate single-rack replication-unit failures and improve database availability.
- Refactored over 1,200 lines of Sharding serialization code to eliminate a critical data-loss defect · Designed robust de/serialization pipelines and multi-format conversion logic that greatly improved data integrity and system reliability.
- Optimized multiple replica-to-replica data transfer operations by using batching · Achieved a **700%** performance improvement offline and **250%** improvement online after optimization code changes.
- Resolved critical recoverability issue in replica-to-replica data transfer operations by ensuring complete transfer of essential recovery-related tables and data between replicas · Improved reliability and system integrity during failure.
- Enhanced error-handling mechanisms throughout the codebase · Improved error handling reliability and diagnosability.

Gecko Robotics — **Pittsburgh, PA**
Software Engineer Intern | **C++, Python, C, CMake, GCP, CI/CD** — **May 2023 – Aug 2023**

- Worked on the Robot Controls team · Wrote code for an asynchronous distributed system in **C++**, **Python**, and **C**.
- Revamped Robot & DAQ emulators · Implemented communications protocol and Client/Server TCP networking.
- Replaced VS build-system with **CMake** · Added calibration support to emulators · Integrated **CI/CD** and **Poetry**.

Carnegie Mellon University - Software Engineering Institute — **Pittsburgh, PA**
DevOps Engineer Intern | **Python, Rust, CI/CD, Docker, Kubernetes** — **May 2022 – May 2023**

- Developed **Python**, **Rust**, and **Bash** code on various NDA projects · Updated and created **CI/CD** pipelines · Created GitLab **REST API** live data visualizations for clients · Used ArgoCD and Helm to deploy **Kubernetes** clusters.
- Built internal analytics tools using **Python**, **Neo4j**, NeoDash, and PageRank-based graph analysis to generate engineering metrics and documentation adopted across multiple teams within the company.

Relevant Academic Work Experience

Georgia Tech Contextual Computing Group **Atlanta, GA**
Graduate Researcher | **AI, ML, Python, Typescript** **June 2025 – July 2026**

- Worked on Nosi, an IDE that combines an encrypted virtual file system and application behavior-logging · Utilizes **Machine Learning** techniques to analyze behavioral data and flag potentially-cheating students.
- Published at **Learning at Scale 2026** · Integrated the IDE into multiple Georgia Tech graduate-level CS courses.
- Created models that can detect the difference between human-coding and code transcription · Implemented tools to help review flagged cheating cases and replay student interactions with the IDE · Developed scripts that automatically generate encrypted repositories for students and decrypt them for the auto-grader.

Georgia Tech **Atlanta, GA**
Graduate Teaching Assistant | **AI, Python, NumPy** **May 2025 – May 2026**

- Taught graduate-level **Artificial Intelligence** · Helped with new plagiarism-detection tooling integration.

Cornell University **Ithaca, NY**
Undergraduate Teaching Assistant | **Python, NumPy, SQL, Java** **Aug 2022 – May 2024**

- Received the **Course Staff Exceptional Service Award** · Taught Databases, Object-Oriented Programming, and Physics · Lectured 400+ students with custom-designed lecture content · Led Exam sessions and other TAs.

Cornell Database Group **Ithaca, NY**
Undergraduate Researcher | **ML, Java, AWS** **Jan 2022 – May 2024**

- Published and presented paper at **VLDB 2023** combining **Reinforcement Learning** and Worst-Case Optimal Joins.
- Worked on upgrading query engine to a **Distributed Architecture** · Created dynamic statistics & RL visualizations.

Publications

Learning at Scale (2026) [https://dl.acm.org/doi/abs/10.1145/3774398.3811574]
Nosi IDE: Securing Coding Assessment Integrity in the Age of LLMs via Process-Driven Analytics

Learning at Scale (2026) [https://dl.acm.org/doi/abs/10.1145/3774398.3811574]
Demonstrating Nosi IDE: Securing Coding Assessment Integrity in the Age of LLMs via Process-Driven Analytics

VLDB (2023) [https://dl.acm.org/doi/abs/10.14778/3611540.3611629]
Demonstrating ADOPT: Adaptively Optimizing Attribute Orders for Worst-Case Optimal Joins via Reinforcement Learning

Awards

Oracle, Database Organization *Mentorship and Leadership Excellence Award*
Recognizes outstanding efforts in mentorship and training of others that go above and beyond the call of duty. Multi-day, multi-recipient efforts beyond the direct organization.

Cornell Bowers CIS *Course Staff Exceptional Service Award*
Faculty-nominated award for exceptional commitment, hard-work, dedication, and exceptional service of a Teaching Assistant during the 2023-2024 academic year.

Relevant Projects

Inference Serving Platform | Personal Project | ***Go, Kubernetes, vLLM, Docker*** | *Github* **April 2026 – Present**

- Building a cloud-hosted inference serving platform for deploying open-source LLMs behind scalable HTTP APIs · Implementing request routing, model container orchestration, autoscaling, and GPU-aware scheduling on Kubernetes.
- Integrating vLLM-based inference workers with asynchronous job queues, health checks, structured logging, and deployment metadata to support reliable model serving across multiple replicas.

Improving 3D Point-Cloud Classification | Deep Learning | ***AI, Python, PyTorch*** | *Github* **Aug 2025 – Dec 2025**

- Re-implemented and upgraded state-of-the-art point cloud classification Deep Neural Networks · Implemented PointNet and PointPillars · Upgraded these models by modifying their architectures, adding Dropout, implementing Data augmentation, and introducing Residual Connections · Improved validation accuracy by 4% and massively reduced overfitting.

Distributed Testing Platform (SPEED) | Masters Project | ***Java*** | *Github* | *BOOM* **Aug 2023 – May 2024**

- 1 of 44 projects selected for BOOM 2024 · Worked on SPEED, Scalable Platform for Efficient Execution of Distributed testing, which is a fault-tolerant system that contains a controller node that orchestrates worker nodes running **JUnit** tests on **Java** code · Aggregated worker node findings to the controller · Displayed test results in the frontend to the user once all tests finish.

SpGEMM Parallelization on GPUs | Parallel Computing | ***CUDA, C++*** | *Github* **Jan 2024 – May 2024**

- Parallelized sparse-matrix multiplication operations using GPUs · Experimented with multiple variations of the algorithm which massively improved performance of the algorithms.